

DebateLab — Learning & Growth Analytics

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jknguyen-portfolio.vercel.app/projects/debatelab-analytics
github.com/ndkn-code/debatelab-analytics

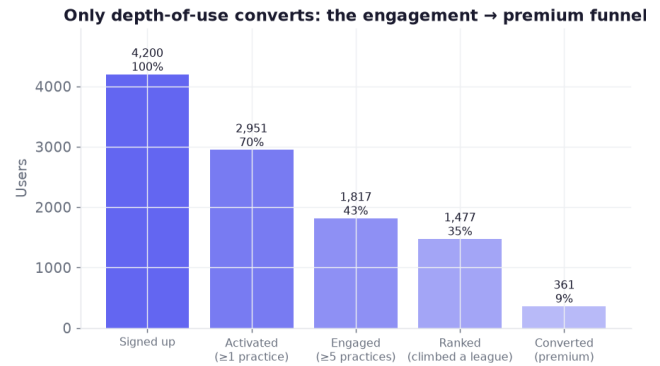
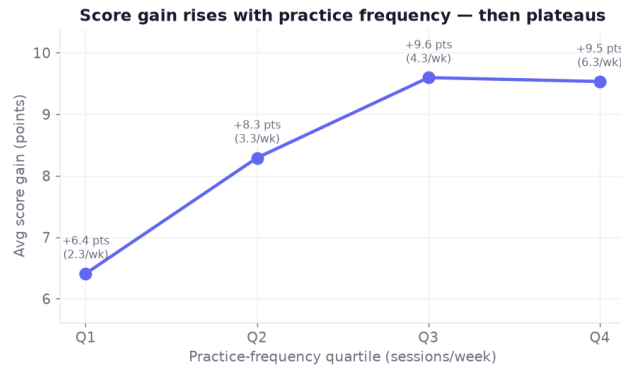
Product analytics (DA + BA) · SQL · Python · Tableau

BUSINESS QUESTION

What makes a learner improve — and what turns a free user into a paying one?

APPROACH

Modeled the product schema and wrote SQL (CTEs, window functions, cohort logic), validated in pandas. 4,200 users / 52K practice sessions (synthetic dataset on the real schema).



KEY FINDINGS

- Practice gains rise with frequency, then plateau past ~4 sessions/week.
- Of 5 scored skills, clarity improves fastest (+19 pts) and rebuttal slowest (+9).
- Engagement funnel: 70% activate, 43% reach 5 practices, 8.6% convert to premium.

RECOMMENDATIONS

- Engineer a 5-practice habit in week 1 — the funnel's biggest leak.
- Concentrate coaching on the slow-moving skills (rebuttal, logic).
- Design streaks around ~4 sessions/week, where gains plateau.

Synthetic dataset modeled on a real production schema (confidential). Full case study & code: jknguyen-portfolio.vercel.app/projects/debatelab-analytics · github.com/ndkn-code/debatelab-analytics